Ankush Gupta
sites.google/ankush-gupta

Academic Statement of Purpose

Purdue University PhD

I want to build AI solutions that are both adaptive and equitable for assisting humans in everyday lives. Across personal experiences, academic work, and industry roles, I have learned how factors such as identity, lived experience, and geographical settings shape people's interactions with technology. These nuances motivate me to evaluate the shortcomings of current AI systems, particularly in affective computing and collaborative AI through behavioral modeling and developing responsible solutions. Over two years, this has driven my work in two research groups and international collaborations in mental health technologies and queer-inclusive language systems, building my foundation for human-centered, socially responsible AI.

Investigating Multi-Source AI Systems: My primary research collaboration with **Prof. Pushpendra Singh (IIIT Delhi)** and **Prof. Mohan Kumar (RIT New York)** is on modelling physiological signals for emotion recognition and mental well-being without compromising user agency. I worked across three paradigms: **evaluating existing solutions**, **developing AI algorithms**, and **conducting user-centered evaluation**. I designed benchmarking and quantitative analysis on model performance tradeoffs. Also collecting multi-modal lab-setting emotional data (physiological signals, text) and developing the first language contrastive pretrained model for physiological signals. The findings were published as co-authored papers at **NeurIPS'24**[1] and **NeurIPS'25**[2], teaching me academic writing, quantitative methods, and experiment design.

Together with co-authors, I performed qualitative analysis with individuals of various backgrounds and mental health specialists to comprehend real-world barriers. Drawing from the themes and feedback from domain-experts, framework of 15 pragmatic guidelines was accepted at **ACM IMWUT'25**[3]. Building on these learnings, I designed the protocol for multi-modal data collection in the wild and the fundamental model for physiological signals (submission planned for January). I've learned about longitudinal collaboration and mixed-methods research from this research project, which has reinforced what I've learned from courses like Explainable AI, HCI, and Deep Learning as well as TAships in Human-Centered AI and Interactive Systems. These experiences deepened my understanding of community-focused technology and my commitment to adaptable, responsible socio-technical solutions.

Research in Responsible Assistive AI: My early work on language-learning tools for children with **Dr. Anmol Srivastava (IIIT Delhi)** expanded my knowledge on cognitive load and proactive interactive design for minority communities. This work was accepted as co-first author at **ACM IndiaHCI'23** and **Springer SCI**[4]. My work on multi-taxonomy recommendation systems and evaluating human bias in existing citations (first author, under-review, **PVLDB 2026**) with **Prof. Mukesh Mohania (IIIT Delhi)** solidified my skills in analytical NLP pipelines and deployment constraints at scale. My industrial roles at **PayGlocal** (adaptive risk modeling for real-time transactions) and **E.ON Digital** (AI orchestration, LLM finetuning) taught me how to build large-scale AI solutions, working on secondary research and real world deployment.

[1] NeurIPS 2024 DOI
[2] NeurIPS 2025 DOI
[3] IMWUT 2025 DOI
[4] Springer Studies in Computational Intelligence DOI

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My **TACL** paper (co-first author) with collaborators from **Carnegie Mellon University** and **The University of Edinburgh**, surveys structural and theoretical gaps affecting LGBTQ+ communities. The work is motivated by observations from my lived experience as a gay individual from a small town. I am currently extending this research to analyze LLM guardrails limitations and data inconsistencies and to develop queer-inclusive language systems. The findings of the LLM evaluations and the construction of queer-inclusive guardrails are in the final stages of the analysis (submission planned for January). Through services as Archival QIA Workshop chair at **NAACL 2025**, I have also learned how to build inclusive research communities. This breadth across sensing, NLP, and community-engaged design has strengthened my foundation in scalable multimodal, responsible AI and developed my core research & collaboration skills.

Purdue University CSE and Future Goals: Through my PhD, I aim to address a fundamental question: **how can we responsibly design multi-source AI agents to understand human experiences while respecting user agency and privacy?** This intersects with the broader inquiry of how Sensing AI systems can adaptively learn from diverse populations in the absence of perpetuating existing inequities or dictating a priori behavioral models. Understanding these questions will ultimately enable me to develop more inclusive and assistive technologies. I am most excited about investigating how such systems can be robust across cultural and demographic diversity and how their design would meaningfully involve the communities they serve.

The interdisciplinary research ecosystem at Purdue offers strong alignment and a holistic space to grow. **Dr. Aniket Bera's** work on multi-sensor human behavior modeling and affective computing, particularly in AI for mental health and **Dr. Ming Yin's** research on human perception and human–AI collaboration through behavioral modeling have shaped my understanding of human-AI collaboration. Their research has influenced the methodologies for NeurIPS and TACL publications. Additionally, the work of **Dr Joseph Campbell** on work multi-modal sensing, spatial computing, and explainable decision-making with heterogeneous data streams has offered great insights and influenced my IMWUT publication. Working under such an academically enriching environment would afford me the methodological rigor and interdisciplinary perspectives which my research on real-time decision-making and responsible AI agents.

I hope to bring perspectives shaped by my lived experiences and research trajectory to Purdue's community. My background in mixed-methods research and fairness-aware design would enable me to contribute to Purdue's vision in developing responsible, socially conscious AI to serve diverse communities. Growth within this environment would bring mentorship and rigor that is required in translating responsible AI principles into scalable solutions, and long-term, in building a research frameworks that centers marginalized communities and advances human well-being.